

## Phase 2

# Intermediate — Index & Study Checklist

Goal: move from single-server fundamentals (Phase 1) to genuinely distributed systems thinking — scaling databases, reasoning about consistency and availability trade-offs, and connecting many services together reliably. Complete chapters in order; each later chapter assumes earlier Phase 1 and Phase 2 chapters.

| #  | Chapter                                                                                               | Time   |
|----|-------------------------------------------------------------------------------------------------------|--------|
| 1  | <b>Scalability Patterns Deep Dive</b><br>Read replicas, connection pooling, scaling reads vs writes   | 1.5 hr |
| 2  | <b>Availability &amp; Reliability</b><br>SPOFs, redundancy, hot vs cold failover, the math of 'nines' | 1.5 hr |
| 3  | <b>Consistency Models</b><br>Strong, eventual, read-your-writes, causal consistency                   | 1.5 hr |
| 4  | <b>CAP Theorem &amp; PACELC</b><br>CP vs AP, and the latency-vs-consistency trade-off beyond CAP      | 1.5 hr |
| 5  | <b>Database Replication</b><br>Single-leader, multi-leader, leaderless; sync vs async; failover       | 1.5 hr |
| 6  | <b>Database Sharding &amp; Partitioning</b><br>Shard keys, hotspots, cross-shard queries, resharding  | 2 hr   |
| 7  | <b>Consistent Hashing</b><br>The hash ring, virtual nodes, why modulo hashing breaks at scale         | 1.5 hr |
| 8  | <b>Message Queues &amp; Async Processing</b><br>Point-to-point vs pub/sub, Kafka, delivery guarantees | 2 hr   |
| 9  | <b>Idempotency &amp; Exactly-Once Delivery</b><br>Idempotency keys, safe retries, webhook handling    | 1.5 hr |
| 10 | <b>Rate Limiting &amp; Throttling</b><br>Token bucket, leaky bucket, fixed vs sliding window          | 1.5 hr |
| 11 | <b>Introduction to Microservices</b><br>Monolith vs microservices, real costs and real benefits       | 1.5 hr |
| 12 | <b>API Gateways &amp; Service Discovery</b><br>Single entry point, service registries, health checks  | 1.5 hr |
| 13 | <b>Search &amp; Indexing Basics</b><br>Inverted index, tokenization, Elasticsearch, sync pipelines    | 1.5 hr |

| #  | Chapter                                                                                                  | Time   |
|----|----------------------------------------------------------------------------------------------------------|--------|
| 14 | <b>Data Modeling for Scale</b><br>Normalization vs denormalization, polyglot persistence                 | 1.5 hr |
| 15 | <b>Phase 2 Recap &amp; Mini System</b><br>Capstone: design a ride-booking notification system end to end | 2 hr   |

**Phase 2 total: ~24.5 hours**

## Study Checklist

Mark each chapter complete only once you can do all three of the following without notes:

- Explain the core concept out loud in under 2 minutes, using the backend-engineer analogy as a bridge.
- Answer all "Common Interview Questions" from memory with a full model-approach structure.
- Complete the practice exercise(s) unaided.

| Done?                               | Chapter                             |
|-------------------------------------|-------------------------------------|
| <input type="checkbox"/> Chapter 1  | Scalability Patterns Deep Dive      |
| <input type="checkbox"/> Chapter 2  | Availability & Reliability          |
| <input type="checkbox"/> Chapter 3  | Consistency Models                  |
| <input type="checkbox"/> Chapter 4  | CAP Theorem & PACELC                |
| <input type="checkbox"/> Chapter 5  | Database Replication                |
| <input type="checkbox"/> Chapter 6  | Database Sharding & Partitioning    |
| <input type="checkbox"/> Chapter 7  | Consistent Hashing                  |
| <input type="checkbox"/> Chapter 8  | Message Queues & Async Processing   |
| <input type="checkbox"/> Chapter 9  | Idempotency & Exactly-Once Delivery |
| <input type="checkbox"/> Chapter 10 | Rate Limiting & Throttling          |
| <input type="checkbox"/> Chapter 11 | Introduction to Microservices       |
| <input type="checkbox"/> Chapter 12 | API Gateways & Service Discovery    |
| <input type="checkbox"/> Chapter 13 | Search & Indexing Basics            |
| <input type="checkbox"/> Chapter 14 | Data Modeling for Scale             |
| <input type="checkbox"/> Chapter 15 | Phase 2 Recap & Mini System         |

## How Phase 2 Chapters Connect

Phase 2 is deliberately ordered so each chapter's mental model supports the next:

- 1 Ch 1 extends Phase 1's scaling story with the specific mechanics of read replicas and connection pooling.
- 2 Ch 2–4 build the reliability/consistency reasoning layer: availability, consistency models, and the CAP/PACELC framework that formalizes their trade-off.
- 3 Ch 5–7 build the data-distribution layer: how replication actually works, how to shard data, and the consistent-hashing technique that makes both scale smoothly.
- 4 Ch 8–10 build the async/safety layer: message queues, the idempotency techniques that make retries safe, and rate limiting to protect shared resources.
- 5 Ch 11–12 build the service-architecture layer: when and how to split into microservices, and how they find and route to each other.
- 6 Ch 13–14 build the data-shape layer: full-text search infrastructure and deliberate normalization/denormalization choices.
- 7 Ch 15 forces you to combine most of the above into one working design — the real test of Phase 2.

### Next step

Once every chapter is checked off and Chapter 15's capstone is solved unaided, you're ready to approve Phase 3 (Advanced) — distributed transactions, security, observability, and the deeper reliability patterns real production systems depend on.